# Parser & SDT

## LL(1) parsing and semantic analysis



# ACME

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ACME designs, develops, and implements computing solutions that turn technical requirements into reliable software.

- Compiler engineering: design and implementation of compilers and language tools.
- Custom software: applications tailored to specific needs and workflows.
- Implementation and integration: testing, deployment, and integration with existing systems.

From compiler design to custom software, ACME builds solutions that work.

# Our team



Ximena

PM



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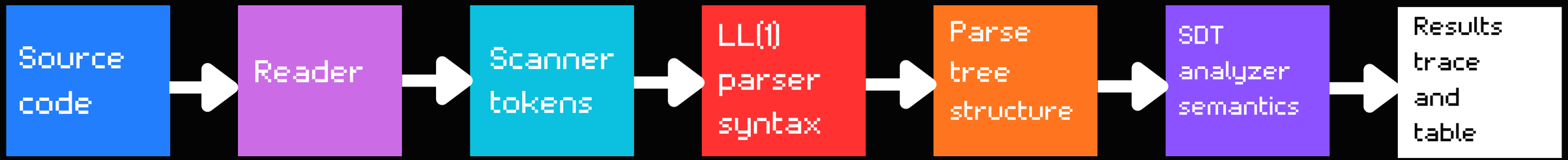
Developer



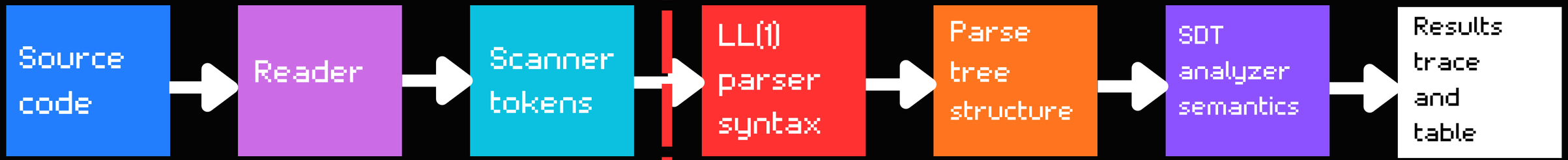
Arely

Writer

# Compiler process



# Compiler process




Lexer

Parser & SDT

# Why LL(1)?

A predictable parser for a compact grammar

One-token lookahead

A stack and parsing table

Linear time parsing O(n)

No backtracking

A step by step trace

Expression precedence

# LL(1) parser



# Expression precedence



# Implementation

The implementation continues to evolve as an interactive web application built with Python and Flask. This stage adds the modules required to construct the parse tree and implement the grammar, parser, and syntax-directed translation (SDT), following the previously established design.

grammar.py

ll1_parser.py

main.py

app.py

# Advantages

Compared with lexer-only or opaque parser implementations:

Transparent parsing.

Clear error diagnosis.

End-to-end analysis.

Detailed symbol tracking.
Testable results.

# Cost

Development

**$195,000**

5 roles, open-source tools

Hosting + reserve

**$32,700**

$12,000 hosting and tools + $20,700 contingency (10%)

Total investment

**$227,700**

MXN · one-time, first year

# After launch

Yearly maintenance

## ~$30,000

MXN per year · about 15% of development cost

- Hosting and domain renewal
- Bug fixes and updates
- Grammar and UI improvements

86% of the budget is talent, with zero license fees: every tool is open source.

# Return on investment

Payback scenarios · assumption: $20,000 MXN per institution license per year

Conservative

**~30 months**

6 licenses per year

Net $90,000 MXN per year

Base

**~16 months**

10 licenses per year

Net $170,000 MXN per year

Optimistic

**~10 months**

15 licenses per year

Net $270,000 MXN per year

Base case: the $227,700 investment is recovered in about 16 months.

Illustrative estimates · net = license revenue − $30,000 yearly maintenance